# Madison Sheridan

madison.sheridan94@gmail.com linkedin.com/in/madison-sheridan 1.916.790.7806


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Dear Hiring Manager,

I am writing to express my interest in **COMPANY**, where I can apply my expertise in computational modeling, numerical analysis, and algorithm development to support innovative engineering solutions.

My Ph.D. research focused on designing and validating high-fidelity numerical methods for simulating nonlinear physical systems. I developed and implemented simulation tools for fluid dynamics, radiation transport, and multiphysics interactions—work that required careful error, stability, and convergence analysis. These skills directly support model verification, performance prediction, and system-level analysis in complex engineering applications.

I bring strong experience in software development and scientific computing. My fluid dynamics models were implemented in C++ within the MFEM finite element library, where I designed and tested invariant-domain preserving methods for compressible flows. I also use Python for error verification, data analysis, and workflow automation on HPC systems, and MATLAB for verification and visualization. I am comfortable in Linux environments and experienced with collaborative software practices such as version control, testing frameworks, and documentation standards.

My technical contributions have led to peer-reviewed publications and technical presentations, demonstrating my ability to deliver reliable, high-performance modeling tools. I thrive in collaborative, interdisciplinary settings—mentoring students, coordinating across applied mathematics and engineering teams, and communicating technical results to diverse audiences.

I am excited by the opportunity to contribute my modeling and computational skills to **COMPANY**'s projects and to help advance its simulation capabilities. I would welcome the chance to bring my technical background, problem-solving ability, and collaborative mindset to your team.

Kind Regards,

Madison Sheridan

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Applicant